Reaction mechanisms (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Evaluate proposed reaction mechanisms and recognise reaction intermediates.
- Distinguish between intermediates and transition states and recognise both in energy profiles of reactions.

In subtopic S1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) you learned how carrying out a chemical reaction is very similar to following a recipe and how the list of measured ingredients is similar to the stoichiometric quantities. The method for carrying out the recipe usually follows the ingredients and is specific for the recipe. Just like we would not dump in everything all at once in order to make cake, a reaction also proceeds in a sequence of steps, known as the reaction mechanism. The reaction mechanism is to a reaction what the instructions are to a recipe.

Individual steps in the mechanism are called elementary steps. For example, we might find out which bonds are broken first, when the transition state is formed and whether the reaction takes place in one or more elementary steps.

Reaction mechanisms are not something we can observe directly, they are based on hypotheses formulated using kinetic and stoichiometric data. The task of the chemist is to carry out experiments to propose a mechanism that best supports the experimental observations, in other words the proposed mechanism is a theoretical mechanism that is consistent with the experimental data. In fact, it is also possible to think of several mechanisms for a single reaction. Working out a mechanism for a reaction can be very challenging and fascinating at the same time. For example, in the reaction:

$$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$$

it has been shown that the reaction has the following possible mechanism:

$$\text{Step 1:} \quad NO_2(g) + \cancel{NO_2(g)} \rightarrow NO(g) + \cancel{NO_3(g)}$$
$$\text{Step 2:} \quad \cancel{NO_3(g)} + CO(g) \rightarrow \cancel{NO_2(g)} + CO_2(g)$$
$$\text{Overall:} \quad NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$$

Figure 1. A reaction mechanism.

More information for figure 1

The image illustrates a reaction mechanism consisting of two steps. Step 1 shows the reaction of NO2 gas with another NO2 molecule, producing NO gas and NO3 gas, with the NO3 being crossed out. Step 2 shows the reaction of NO3 gas with CO gas, producing NO2 gas and CO2 gas, with the NO2 being crossed out. The overall reaction, shown at the bottom, simplifies to NO2 gas reacting with CO gas to produce NO gas and CO2 gas. Cancelled species NO3 and NO2 are shown crossed out in the intermediate steps, indicating they are intermediates and are not present in the overall reaction.

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The overall reaction is a sum of the elementary steps and must be identical to the balanced equation. The overall reaction is obtained by adding the elementary steps together and cancelling species that exist on both sides of the reactions.

Notice that that one molecule, NO_3, appears in both steps but does not appear in the overall reaction (**Figure 1**). In this case NO_3 is a reaction intermediate. A reaction intermediate is a temporary molecule produced in one step and subsequently consumed in the next. An intermediate is typically unstable, so they may not be represented by formulas that are familiar to you.

The rate determining step

The rate determining step can be compared to a series of funnels (**Figure 2**). It is the funnel with the smallest diameter that controls how fast the liquid flows through, whether it is placed first or last. In the same way, in a chemical reaction occurring in several steps, it is the slowest step that determines the rate of the reaction.

Figure 2. The rate determining step can be compared to a series of funnels.

More information for figure 2

The image is an illustration depicting three vertical arrangements of funnels with a blue liquid being poured into the top funnel of each set. Each column has a series of funnels stacked vertically. In each set, the diameter of one of the funnels is notably smaller, illustrating the concept of the rate-determining step. The flow of liquid is continuous from the top to the bottom funnel, eventually collecting in a flask at the base of each arrangement. This visual is used to explain how the narrowest point (smallest funnel) controls the overall rate of flow, similar to how the slowest step dictates the rate of a multi-step chemical reaction.

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For example in the reaction between nitrogen dioxide and carbon monoxide the rate determining step is the first step.

Step 1:

$$NO_2(g) + NO_2(g) \xrightarrow{\text{slow}} NO(g) + NO_3(g)$$

Step 2:

$$NO_3(g) + CO(g) \xrightarrow{\text{fast}} NO_2(g) + CO_2(g)$$

Step 1 limits the rate of reaction, and thus will determine the rate.

International Mindedness

Plastic pollution has become one of the major environmental issues threatening the planet today. The development of biodegradable plastics has been a positive step towards more sustainable solutions. Biodegradable plastics, as the name suggests, breakdown and recycle the elements they contain back into the soil or aquatic environment. Research into plastics involves studying the reaction mechanisms involved in the production and degradation process. The reaction mechanism gives details of how different atoms are reassembled during the process. These reactions can then be manipulated by changing reaction conditions and shuffling atoms, to obtain molecules with desirable properties, like easy decomposition under different conditions, strength and durability.

Worked example 1

A proposed reaction mechanism has the following steps:

Step 1 Fast:

$$2N_2O \rightarrow N_4O_2$$

Step 2 Slow:

$$N_4O_2 \rightarrow 2N_2 + O_2$$

1. Write the overall equation.
2. Which step is the rate determining step?
3. Identify the intermediate.

1. Combine the equations and cancel the intermediate
   $2N_2O \rightarrow 2N_2 + O_2$
2. Step 2, it states it is Slow whereas Step 1 states it is Fast.
3. Intermediate N_4O_2

Worked example 2

Propose two possible reaction mechanisms for the given reaction.

$$2A + B \rightarrow A_2B$$

Step 1: A + B → AB

Step 2: AB + A → A_2B

Overall: 2A + B → A_2B

Notice AB is an intermediate in this reaction, which is produced in the first step and consumed in the next.

or

Step1: A + A → A_2

Step 2: A_2 + B→ A_2B

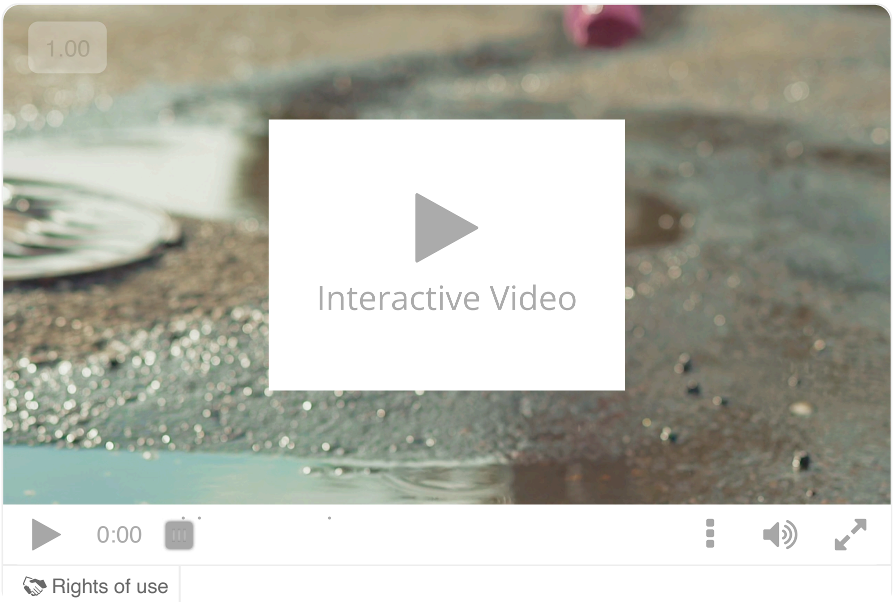
Overall: 2A + B → A_2B

A_2 here is an intermediate.

Transition states and reaction intermediates

In section R2.2.4–5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/) you learned about transition states. It is important to distinguish between transition states and intermediates in order to draw reaction mechanisms.

Imagine a walk through a park (**Interactive 1**), if you decide to jump over a series of puddles along a path, when you are in the middle of your jump, you are neither walking nor touching the ground, this is your transition state. You cannot freeze mid-air as you hover over a puddle at your transition state. The same thing applies to transition states in reactions, they are unstable and quickly revert into products. If then, upon landing you pause momentarily on one leg before jumping over the next puddle, this is your intermediate stage before you actually land on two feet and start walking again. You can stop at your intermediate, you might wiggle a little but are generally quite stable. A reaction intermediate is the same. It is relatively stable and in some cases stable enough to be isolated.



Interactive 1. Jumping Over Puddles.

More information for interactive 1

An interactive video displays a child wearing rain boots and jumping over puddles to demonstrate the transition state and intermediate state of a reaction.
As the video progresses, it freezes when the child is in mid-air above the first puddle and a text appears as "the transition state". When the child lands in the puddle, this stage is referred to as "the intermediate state".
As the video continues further, the child takes two steps forward and jumps again. The video freezes mid-air to indicate another transition state. Then the child lands on the second puddle, causing the water to splash onto their legs and outside the puddle.

Transition states

The transition state and reaction intermediates can be represented in energy profile diagrams. For a single step reaction the transition state can be represented as shown in **Figure 3**. The transition state occupies the highest point on the reaction profile diagram, shown as the 'peak' of the hill.

- The transition state is not the reactant or product. It resembles both as only some bonds in the reactant are broken and some other bonds are formed.
- The transition state exists for a very short period of time and hence cannot be isolated from the reaction mixture.

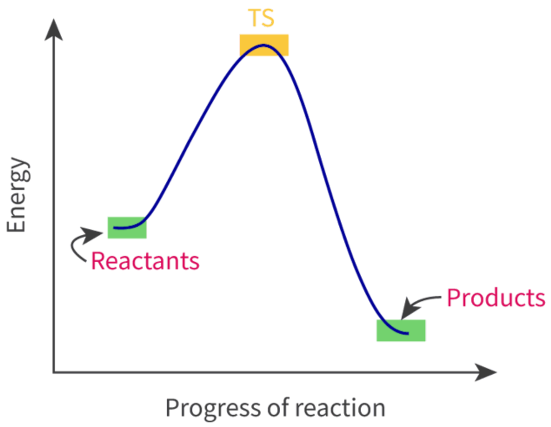


Figure 3. The energy profile diagram showing the transition state.

More information for figure 3

The image is a graph illustrating an energy profile diagram for a chemical reaction. The X-axis represents the 'Progress of reaction' while the Y-axis represents 'Energy.' The diagram features a curve that starts at the energy level of the reactants, rises to a peak indicating the transition state (TS), and then declines to the energy level of the products. The curve begins with reactants labeled at a lower energy point, ascends to the top where the transition state is labeled 'TS,' and descends to a lower energy point labeled 'Products.' The transition state marks the highest energy level on the graph. Arrow indicators show the progression from reactants to products, emphasizing the peak as a critical point in the reaction process.

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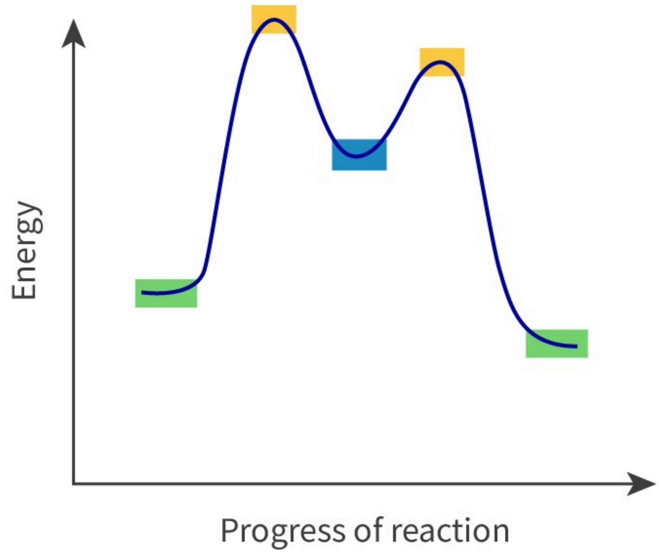
Intermediates

Intermediates are species with lower energy when compared to transition states. However, they are still quite unstable and in most reactions they cannot be isolated. Not all reactions have intermediates. There are plenty of single-step reactions that do not have intermediates.

Interactive 2 shows a two-step reaction. Can you identify where the transition states and intermediates would be on the energy profile diagram? Use the slider along the bottom of the energy profile to reveal the answer.



Interactive 2. The Energy Profile Diagram Showing the Energy of the Intermediate and Transition States.

More information for interactive 2

This interactive slide graph illustrates the energy changes of reactants, intermediates, and products, along with their energy transition states throughout chemical reactions. The slider starts with a graph with the x-axis labeled as the progress of the reaction and the y-axis labeled as energy. Initially, the curve begins at a low energy point. As the curve rises, it reaches its first peak and after reaching the first peak, the curve decreases in energy, forming a low valley. It then rises again, forming a second peak. This peak again decreases, leading to a very low point.
As the slider progress curve of the graph shows labelled positions. The first low energy point is labelled as reactants on the left side of the curve. The curve's first peak is labelled as TS1, which is the first transition state. The curve decreasing in energy is labelled as intermediate. The second peak of the rising curve is labelled as TS2 that is the second transition state. Finally, the curve ends with a low-energy point and is marked as Products.
This interactive demonstrates how energy changes during a two-step reaction and explains key concepts such as transition states, intermediates, and overall reaction energy flow.

Table 1 shows a comparison between transition states and intermediates.

Table 1. Summary of the differences between transition states and intermediates.

Transition states	Intermediates
High energy state	Lies between two high energy stat
Have a very short lifetime so are unstable and cannot be isolated	Are usually unstable and in most reac cannot be isolated*
Every reaction has a transition state	Exist only in two or more step reacti
Some reactant bonds are broken and some product bonds are formed	All of the reactant bonds are broken to the intermediate product

*In general, reaction intermediates cannot isolated. However, there may be certain reactions where it is possible.

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Inquirer
  - Thinker
  - Reflective
- **Approaches to learning:**
  - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
  - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
  - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

In this activity you will work in groups of five to ten.

Materials

- Two sets of ten unlabelled ping pong balls
- Two spoons or spatulas

Activity

This activity is a relay race between two groups of students.

- Divide the class into two groups and sit each group in a line with a basket of ten ping pong balls at one end of each line. The race will involve passing the balls from one basket to another empty basket at the other end. The team that passes all ten balls from one basket to the other first, wins.
- Most of you will use your hands. One person in each team must use a spoon or spatula and one more person in each team should stand up when he receives the ball, jump twice and turn in a circle before he passes the ball on.

Rule

If the person you are trying to pass the ball to already has a ball, hold on to yours. A person can only have one ball at a time.

In your teams discuss the following questions:

1. What was the main factor that influenced how fast the balls were passed?
2. Based on what you have learned from this activity, explain how this can be applied to reaction mechanisms and the rate of reaction.